

Ain Foundry vEIR Technical Architecture

Reference Architecture, Data Boundaries, Interfaces, Security Controls, and Verification Plan



Provider-neutral

owned brain and memory

Venture-isolated

authorization at every retrieval

Target <800 ms

warm P50 first audible response

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Audience	Capstone engineers, University technical reviewers, product owners, security and privacy stakeholders, and future implementation teams
Date	September 2026

Working name: Ain Foundry Virtual Entrepreneur-in-Residence (vEIR). This document proposes a human-led, AI-amplified capability; it does not authorize deployment or the use of any person's likeness.

REFERENCE DESIGN

1. Architecture decisions at a glance

Recommended V1 architecture

Build a single, policy-controlled conversational orchestrator with venture-scoped retrieval and memory. Use a replaceable real-time avatar provider for the body layer. Do not place institutional memory, EIR doctrine, or safety decisions inside the avatar vendor's proprietary conversation stack.

Decision	Selected direction	Rationale
System boundary	Ain owns identity mapping, orchestration, RAG, memory, policies, evaluation, and audit.	These are the durable program assets and highest-risk trust functions.
Avatar integration	Provider adapter; Avatar Only / Lite-style path for V1 when approved.	Fast route to real-time embodiment without surrendering the vEIR brain. [S7]
Agent design	One orchestrator plus deterministic policy, authorization, retrieval, and response validators.	Easier to test and audit than an unconstrained multi-agent swarm.
Memory	Explicit typed objects; AI proposes durable updates, authorized users approve material memory.	Prevents inference from silently becoming venture fact.
Isolation	Every request and data row scoped by user, venture, role, EIR, and data class.	Avoids relying on prompts to enforce confidentiality.
Persona	Expertise, judgment, communication style, voice, and visual likeness stored and governed separately.	Supports consent, testing, portability, and partial retirement.
High-risk advice	Educational framing plus mandatory human escalation.	The platform must not impersonate professional or personal approval.
Deployment	Use University-approved identity, cloud, and vendors; retain a local development and synthetic-data mode.	Institutional approval status is a deployment prerequisite, not an engineering afterthought.

Architectural quality attributes

- **Confidentiality:** zero unauthorized cross-venture retrieval, disclosure, export, support access, or analytics exposure.
- **Traceability:** every response can identify sources, memory objects, policies, model and persona versions, and escalation outcome.
- **Correctability:** founders and EIRs can correct eligible memory and doctrine without retraining a foundation model.
- **Portability:** voice, avatar, and model providers can change without rebuilding the venture workspace.
- **Graceful degradation:** text and evidence remain available if video, voice, or an external provider fails.
- **Accessibility:** captions, transcript, keyboard operation, text-only mode, and understandable disclosure are first-class paths.

C4 - CONTEXT

2. System context and actors

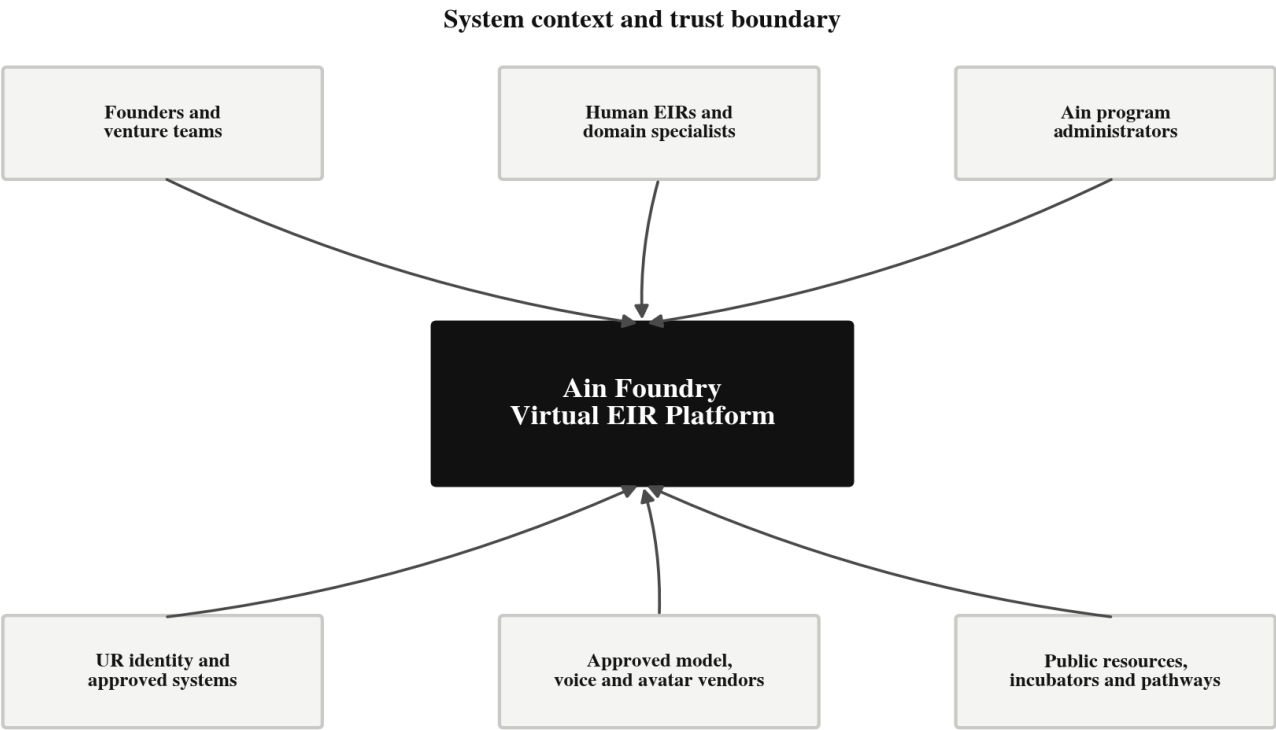


Figure 1. External actors and services around the vEIR trust boundary.

Actor / system	Inputs	Outputs	Trust rule
Founder	Speech, text, venture artifacts, corrections, sharing consent.	Advice, evidence, memory view, recap, escalation request.	Can access only authorized ventures and approved program resources.
Human EIR	Doctrine, cases, persona approvals, corrections, escalation responses.	Review samples, issue packets, persona metrics.	Access follows assignment and minimum-necessary sharing.
Ain operator	Enrollment, EIR mapping, resources, policy, routing, suspension.	Portfolio-safe operations and quality signals.	Cannot retrieve raw venture content without an authorized role and purpose.
UR identity / systems	Authentication, groups, approved resources, possibly calendars or document systems.	Identity claims and authorized content.	Least privilege; connector-specific consent and audit.
Model / speech / avatar providers	Minimized session payloads and approved persona assets.	Transcription, reasoning, audio, video stream.	No training or retention beyond approved terms; contract and technical controls required.
External resources	Public or licensed venture information.	Current pathways, rules, opportunities, sources.	Provenance, freshness, licensing, and allow-list controls.

PUBLIC PROFILE IS NOT RUNTIME AUTHORITY

2A. EIR expertise registry and assignment boundary

The public Ain directory currently exposes seven EIR profiles. The product should ingest those profiles only as **candidate expertise metadata**. Runtime authority requires an active program assignment, scoped persona consent, approved doctrine release, and explicit venture access. [S3]

Listed EIR	Candidate expertise tags	Possible routing intent
John Schloff	Growth, marketing leadership, product management, emerging technology	Go-to-market design; customer process; product portfolio; traditional-market entry
David Mammano	Entrepreneurship, leadership, education, motivation	Founder leadership; narrative; workshop-style coaching; resilience and communication
Kate Washington	Finance, operations, economic development, global growth	Operating model; financial discipline; public-private partnerships; international expansion
Roberto Colangelo	Strategy, entrepreneurship, global marketing, biotech, energy	Venture strategy; commercialization; emerging markets; fundraising and government modernization
Scott Walker	Finance, operations, capital projects, sales and investor relations	Startup finance; operating cadence; capital planning; sales execution; investor readiness
Dina Uzri	Science-to-market commercialization, biotech, product and business development	Research translation; value proposition; life-science pathway; fundraising and strategic planning
Truptesh H. Kothari	Clinical program building, health-system operations, multidisciplinary growth	Clinical innovation; healthcare delivery; stakeholder alignment; quality and scalable programs

Assignment object

```
{ "assignment_id": "...", "venture_id": "...", "eir_id": "...", "persona_release_id": "...", "allowed_modes": ["diagnose","coach","rehearse"], "data_class_ceiling": "venture_private", "starts_at": "...", "expires_at": "...", "status": "active" }
```

Hard guard

A matching score can recommend an EIR; it cannot activate a persona or grant data access. The session service must resolve an approved assignment server-side on every session creation.

C4 - CONTAINERS

3. Logical container architecture

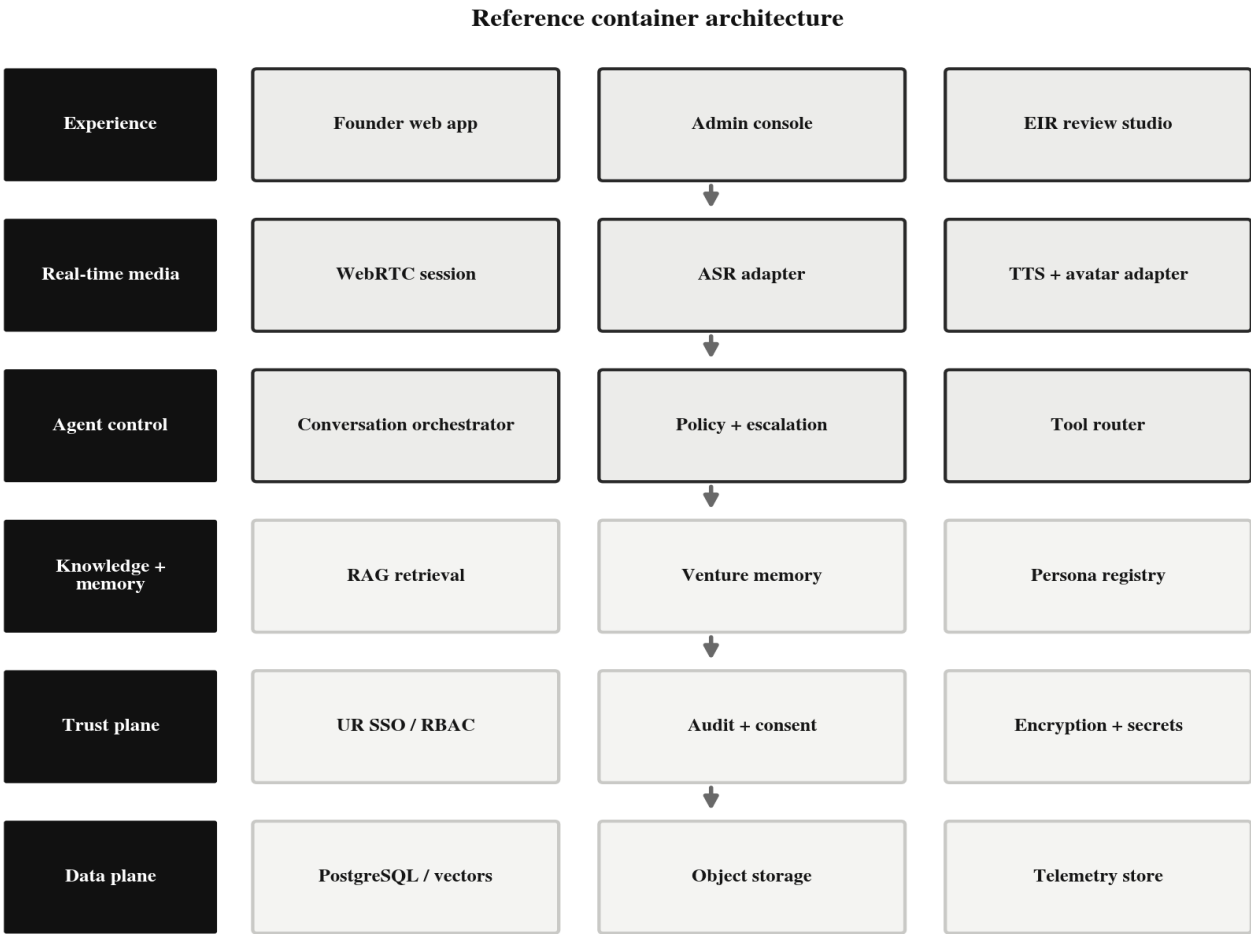


Figure 2. Layered reference architecture. Trust controls are cross-cutting even though shown as a dedicated plane.

Concrete implementation profile for a capstone prototype

Container	Suggested implementation	Responsibility
Founder web app	Next.js App Router + TypeScript; React Server Components where useful; WebRTC SDK; captions and text fallback.	Session UI, microphone / camera controls, evidence and memory panels, recap, escalation consent. [S19]
Admin and EIR studio	Same TypeScript application with role-gated nested routes and shared design system.	Enrollment, persona versions, corpus, scenario tests, sample review, incidents, suspension.
API / session gateway	TypeScript service behind an approved edge gateway, WAF, quotas, and short-lived token issuer.	BFF contracts, authentication enforcement, media-room tokens, correlation IDs, abuse protection.
Realtime orchestrator	TypeScript / Node.js event loop; stateless regional warm workers; WebRTC session events.	Endpointing, barge-in, incremental response planning, policy gates, tool routing, streaming output.
Intelligence services	Python FastAPI + Pydantic for retrieval, memory, ingestion, evaluation, and model experimentation.	Hybrid search, evidence assembly, typed memory proposals, benchmarks, offline jobs.
Retrieval and memory	PostgreSQL + pgvector + full-text search + row-level security; object storage for artifacts.	Authorized hybrid retrieval, typed memory, versioning, provenance, deletion workflow. [S22]
Durable orchestration	Temporal TypeScript / Python workflows outside the audio hot path.	Ingestion, summarization, review queues, deletion, notifications, retries, and long-running approvals. [S23]
Media adapters	LiveKit / WebRTC transport plus typed adapters for ASR, speech-to-speech or TTS, and avatar providers.	Nearest-edge media, vendor normalization, minimized payloads, timeouts, circuit breakers, graceful fallback. [S20][S21]
Observability	OpenTelemetry traces, metrics, structured security events, and privacy-aware replay metadata.	Latency, errors, retrieval lineage, cost, quality samples, incident evidence without raw-content logging by default.

Capstone implementation assumptions

- Begin as a modular monolith with explicit service interfaces; split services only when scaling, security, or release ownership requires it.
- Keep the conversation service stateless. Durable session, evidence, memory, consent, and audit objects belong in authoritative stores.
- Use asynchronous workers for ingestion, summarization, evaluation, deletion, and notification; keep the live turn path narrowly synchronous.
- Represent model, prompt, policy, persona, corpus, and vendor configuration as versioned release inputs attached to every evaluated session.
- Deploy text-first and synthetic-data modes before connecting approved voice, visual likeness, or confidential venture artifacts.

STREAMING SEQUENCE

4. Real-time conversation path

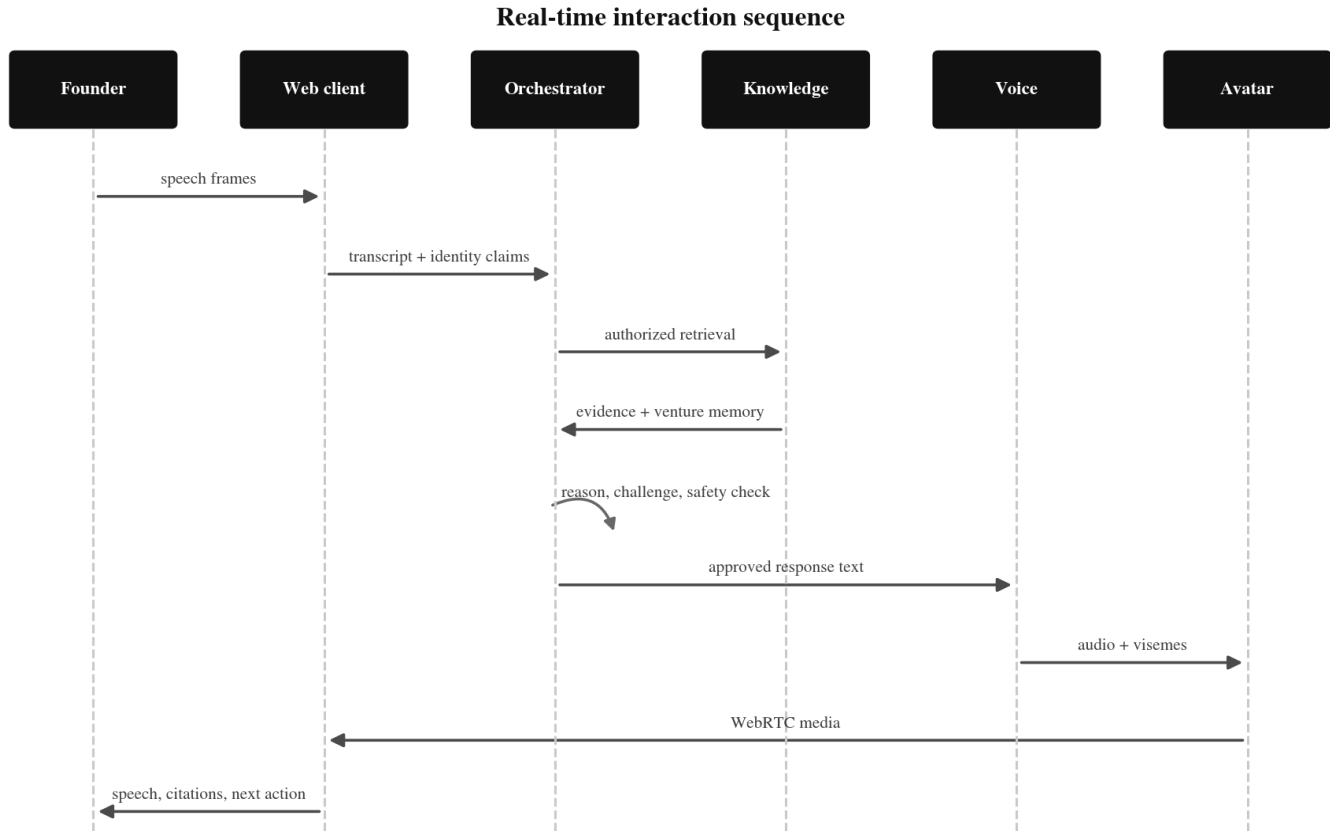


Figure 3. Proposed sequence for voice input through grounded reasoning to synchronized avatar output.

Turn pipeline

1. Establish an authenticated session token containing user, venture, role, assigned EIR, persona version, data-policy version, and expiration.
2. Stream audio to the ASR adapter; preserve timestamps and interruption events. Raw audio is not stored by default.
3. Classify intent, topic risk, requested action, and required data classes before retrieval.
4. Authorize each retrieval independently; apply venture, role, source, and data-class filters at the query and row level.
5. Assemble a bounded context containing the EIR doctrine, relevant venture memory, approved sources, and explicit conflicts or missing evidence.
6. Generate a response plan, run safety and grounding checks, then stream only approved text to TTS and the avatar body.
7. Render synchronized audio / video, live captions, evidence markers, and a visible synthetic-identity indicator.
8. Create a recap and candidate memory updates; commit material durable memory only after the applicable approval rule.

Interruption and failure behavior

Failure	Immediate behavior	Fallback
Avatar stream fails	Stop video, preserve session state, disclose degraded mode.	Continue with audio or text plus citations.
TTS fails	Do not replay stale audio or substitute an unapproved voice.	Text and captions only.
ASR confidence is low	Show uncertain transcript and ask for confirmation.	Text input or push-to-talk retry.
Retrieval fails	Do not answer from persona confidence alone.	State unavailable evidence, offer general framework, or escalate.
Model timeout	Cancel downstream media and preserve idempotent turn state.	Retry once through approved fallback or return a transparent failure.
Authorization changes	Terminate access at the next policy check and invalidate active tokens.	Require fresh authentication and assignment.

Recovery design requirements

- Make every turn idempotent so retries cannot duplicate speech, memory proposals, escalation packets, or billable media sessions.
- Persist a minimal state checkpoint before external calls and cancel downstream media immediately when authorization or policy changes.
- Exercise avatar, speech, model, retrieval, and network failures independently; verify that text, citations, and human routing remain available.
- Expose the degraded mode to the founder and telemetry stream; never disguise a fallback as normal operation.
- Define recovery-time, data-loss, and incident-notification objectives before a controlled pilot, then test them with timed drills.

AGENT CONTROL PLANE

5. Conversation orchestrator

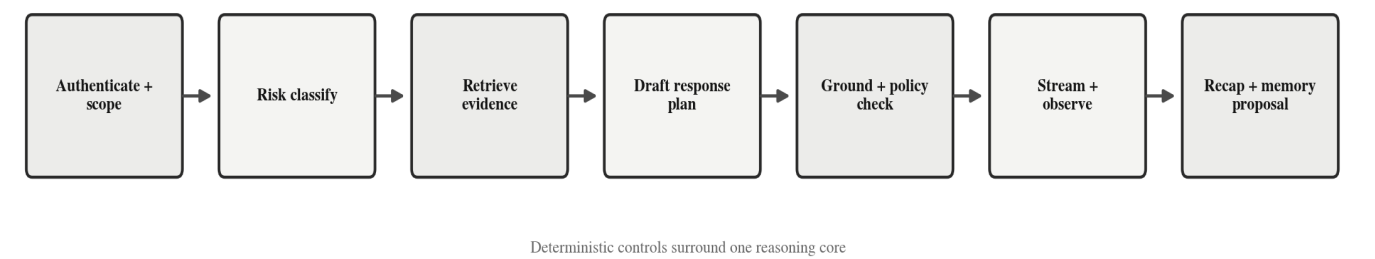


Figure 4. V1 favors a testable orchestrator over autonomous agent proliferation.

Internal state machine

State	Allowed transition	Guard
NEW	AUTHENTICATED or DENIED	Identity valid; session purpose and venture scope supplied.
AUTHENTICATED	LISTENING	Persona active; consent and policy versions current.
LISTENING	UNDERSTANDING	ASR confidence adequate or founder confirms transcript.
UNDERSTANDING	RETRIEVING, CLARIFYING, or ESCALATING	Intent and risk classification complete.
RETRIEVING	REASONING	All records satisfy independent authorization filters.
REASONING	RESPONDING, CLARIFYING, or ESCALATING	Grounding, disclosure, and output-policy validators pass.
RESPONDING	LISTENING, SUMMARIZING, or ENDED	Media sequence matches approved response text.
SUMMARIZING	MEMORY_REVIEW or ENDED	Recap distinguishes user statements, retrieved facts, and model inferences.
MEMORY_REVIEW	COMMITTED or DISCARDED	Approver and object-level rule satisfied.
Any active state	SUSPENDED	Revoked persona, security event, expired access, or operator action.

Response contract

The reasoning service returns a typed object before any text is converted to voice:

```
{ "mode": "diagnose|challenge|coach|teach|rehearse|route", "answer": "...", "claims": [{"text": "...", "source_ids": ["..."]}], "uncertainty": [{"item": "...", "level": "low|medium|high"}], "next_actions": [{"owner": "...", "action": "...", "target": "...}], "escalation": {"required": true, "category": "IP", "reason": "..."}, "memory_candidates": [...] }
```

GROUNDED CONTINUITY

6. Knowledge and memory architecture

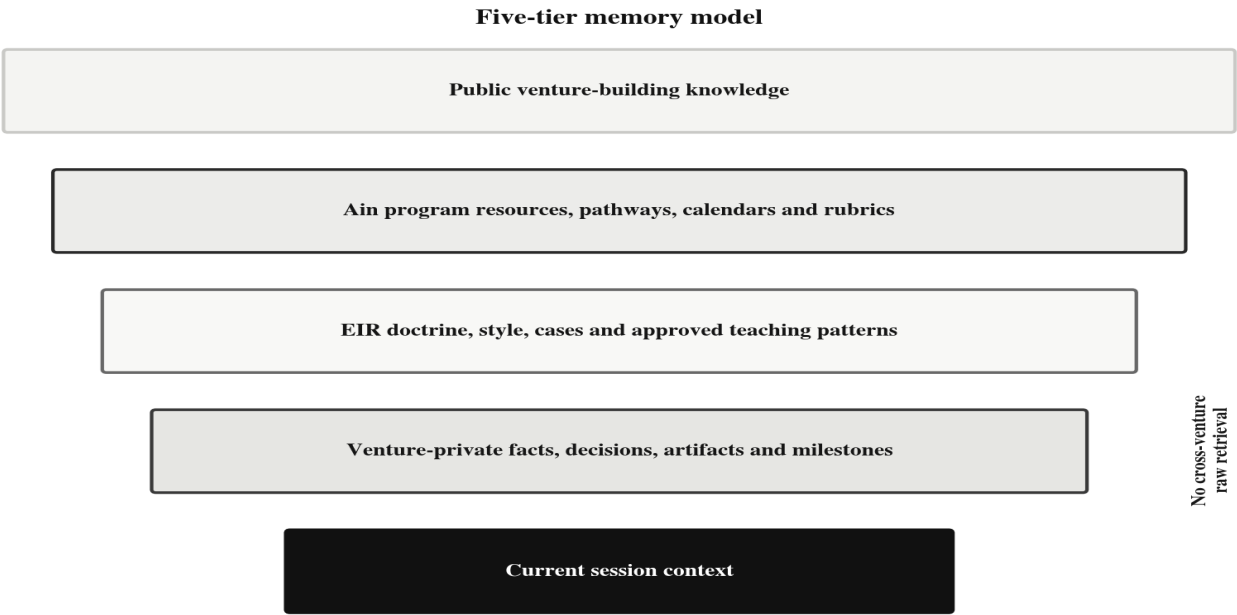


Figure 5. Retrieval order and governance differ by memory tier.

Store	Canonical content	Retrieval rule	Write rule
Public knowledge	Public venture frameworks and current external resources.	Allow-listed source, freshness, license, and relevance filters.	Curator ingest plus scheduled refresh.
Ain program	Programs, mentors, EIRs, competitions, templates, routing and events.	Program-wide unless marked restricted.	Named Ain content owner approval.
Persona registry	Doctrine, cases, style card, consent, voice / avatar references, test status.	Only active assigned persona and approved content.	EIR or delegated persona curator; versioned approval.
Venture memory	Facts, hypotheses, decisions, experiments, milestones, artifacts, corrections.	Exact venture plus object-level permissions; deny missing scope.	Authorized user or AI proposal followed by rule-based approval.
Session store	Turn state, transcript, evidence links, feedback, quality events.	Session participants and authorized reviewers.	Automated; retention minimized and policy-driven.
Pattern library	Reviewed and de-identified program lessons.	Program-wide with provenance and expiry.	Two-step de-identification and publication approval.

Hybrid retrieval algorithm

1. Resolve the request scope and generate a query plan by store; never perform an unscoped global vector search.
2. Apply relational authorization filters before semantic ranking, not as a post-processing filter.
3. Combine lexical, semantic, recency, source-authority, and venture-stage scores.
4. Retrieve contradictory and superseded records when they materially affect the answer.
5. Return compact evidence with object IDs, source IDs, dates, owners, permissions, and confidence.
6. Reject the answer or narrow it when evidence coverage falls below the policy threshold.

TYPED, AUDITABLE OBJECTS

7. Core data model

Entity	Key fields	Security / lifecycle fields
User	user_id, identity_provider_id, display_name, affiliation	status, roles, policy_ack_version, last_access_review
Venture	venture_id, name, stage, domain, program_cohort	classification, status, owner_ids, retention_policy_id
VentureMembership	venture_id, user_id, role	granted_by, granted_at, expires_at, revoked_at
EIRPersona	persona_id, eir_id, doctrine_version, style_version, voice_ref, avatar_ref	consent_id, allowed_uses, audience, active, expires_at, approver
KnowledgeSource	source_id, title, owner, URI, checksum, extracted_text_ref	classification, allowed_personas, freshness_at, license, status
MemoryObject	memory_id, venture_id, type, content, source_refs, confidence	visibility, created_by, approved_by, version, supersedes, expires_at
Session	session_id, venture_id, persona_id, participants, mode, start / end	policy_version, model_version, vendor_refs, retention_state
Turn	turn_id, session_id, transcript_ref, response_ref, claim_refs	risk_label, validator_results, latency, cost, feedback
Escalation	escalation_id, venture_id, category, packet_ref, assignee, priority	sharing_consent, status, due_at, resolved_by, resolution_ref
Consent	consent_id, subject_id, asset_types, approved_uses, audiences	provider_terms_version, granted_at, expires_at, revoked_at, evidence_ref
AuditEvent	event_id, actor, action, object_type, object_id, request_id	timestamp, outcome, policy_reason, immutable_hash

Row-level authorization invariant

A data record is retrievable only when the authenticated user is an active member of the record's venture or holds a separately authorized review role; the requested persona is active and assigned; the record's data class is permitted for the session purpose; and no revocation, legal hold, retention, or incident policy blocks access. The application must not trust a venture identifier supplied by the browser without re-deriving authorization server-side.

Illustrative policy test

DENY retrieval when request.user_id is not an active VentureMembership for object.venture_id; DENY when object.classification exceeds session.allowed_data_class; DENY when persona.active is false; DENY when source.allowed_personas excludes session.persona_id; otherwise ALLOW and record the policy decision without logging raw object content.

DEFENSE IN DEPTH

8. Isolation, security, and privacy controls

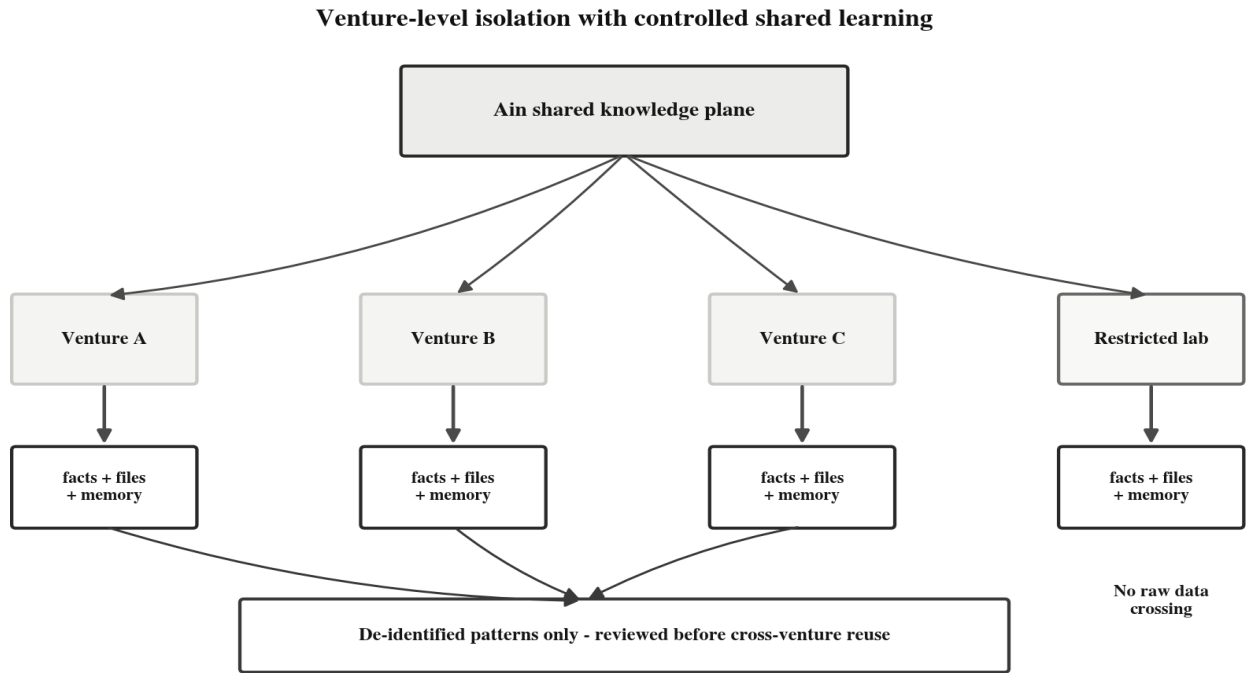


Figure 6. Logical isolation must also cover logs, analytics, support tools, backups, model payloads, and exports.

Control family	Required V1 control	Verification
Identity	Approved authentication; short-lived tokens; phishing-resistant MFA where available; no shared accounts.	Token, session, expiry, role-change, and disabled-user tests.
Authorization	Deny-by-default service checks and database row-level security; object-level connector scopes.	Positive and negative matrix across user, venture, role, persona, and class.
Encryption	TLS in transit; managed encryption at rest; keys and secrets outside source code; rotation path.	Configuration review and key / secret scan.
Data minimization	Send only necessary context to each external provider; raw audio / video storage off by default.	Payload inspection and retention audit.
Logging	Structured metadata and security events; raw prompts and documents excluded from general logs.	Log sampling, support-tool inspection, redaction tests.
Network	Private service paths where supported; outbound allow list; WAF; rate and abuse limits.	Architecture review and egress tests.
Vendor	No-training, retention, subprocessor, breach, deletion, likeness, and exit terms reviewed.	Contract / security checklist tied to technical configuration.
Incident	Kill switch, persona suspension, token revocation, evidence preservation, notification workflow.	Tabletop exercise and timed drill.
Backup / deletion	Encrypted backups inherit scope and retention; deletion propagates and is verified.	Restore test and deletion-certificate workflow.

University data rule

The prototype should begin with public, synthetic, or expressly authorized low-risk data. Use of program-internal, venture-private, FERPA-related, health, regulated, proprietary, or research data requires classification and the applicable University approval. UR guidance warns against entering non-public or sensitive data into unapproved GenAI services and identifies FERPA education records and other protected classes. [S11][S12][S13]

SYNTHETIC IDENTITY SUBSYSTEM

9. EIR persona and likeness pipeline

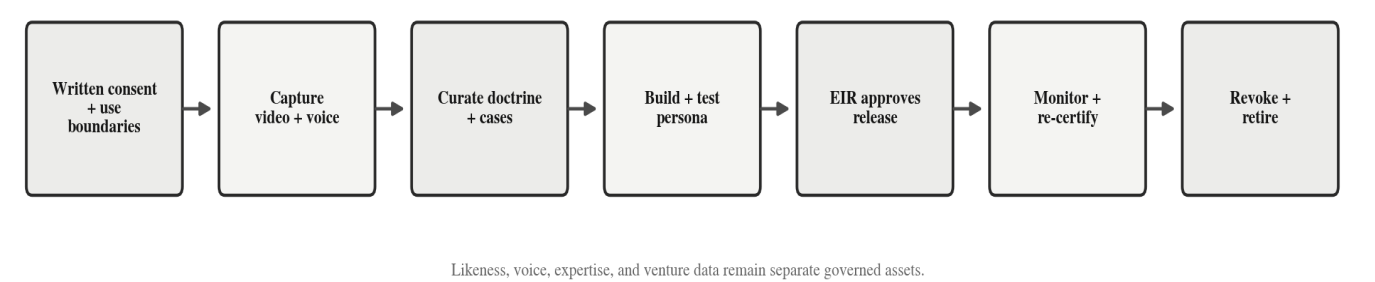


Figure 7. Persona assets are versioned, approved, and revocable; a provider's model is never the sole governance record.

Capture package

Asset	Minimum content	Storage / use boundary
Consent record	Identity, assets, purposes, audiences, platforms, term, withdrawal, provider disclosures.	Governance store; authoritative regardless of vendor UI.
Video capture	Provider-compliant listening and speaking footage; neutral and representative movements.	Encrypted restricted asset; upload only to approved provider and approved use.
Voice capture	Consented script and speech samples sufficient for the selected voice route.	Separate voice reference and model; independent retirement control.
Doctrine interview	Frameworks, stage logic, evidence standards, typical questions, referrals, red lines.	Curated text knowledge; EIR review required.
Decision cases	Representative scenarios, preferred reasoning, counterexamples, escalation choices.	Evaluation and retrieval assets; versioned gold standard.
Style card	Directness, warmth, vocabulary, pacing, teaching behavior, unacceptable imitation.	Prompt / policy input, not a claim of identity.

Release criteria

- The EIR approves the current persona card, representative sessions, disclosure language, and escalation doctrine.
- The provider and University requirements for consent, restricted access, retention, disclosure, and feedback are satisfied. Microsoft custom voice and avatar services are limited-access and require explicit consent and synthetic disclosure. [S5]
- Voice and visual output do not imply that the human is live; captions and a text path work without embodiment.
- The team can disable one asset layer - visual, voice, style, doctrine, or entire persona - without corrupting venture data.

BUILD VERSUS BUY

10. Avatar and speech provider abstraction

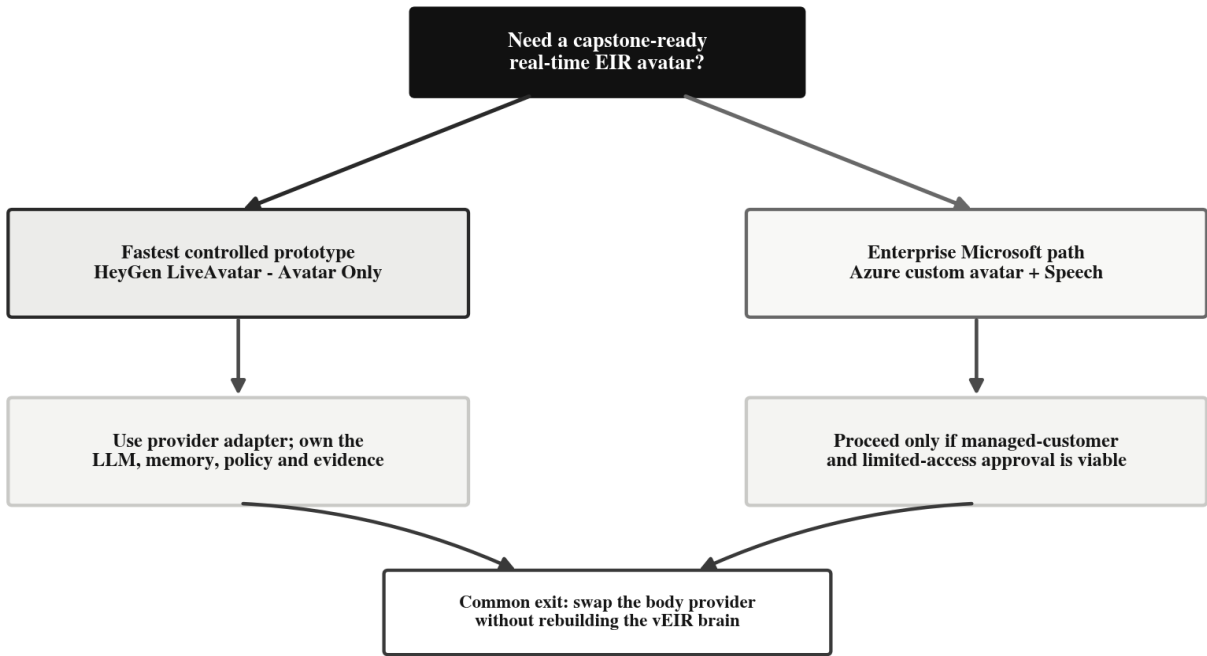


Figure 8. Recommended path: own the vEIR brain and use a swappable provider for embodiment.

Criterion	HeyGen LiveAvatar - Avatar Only	Azure custom avatar + Speech	Architecture implication
Prototype speed	Purpose-built real-time avatar; custom LiveAvatar capture described as one continuous two-minute video. [S8]	Real-time avatar via Speech SDK / WebRTC; custom video avatar requires training footage. [S6]	Keep both behind one session adapter.
Bring your own brain	Avatar Only mode expects customer-provided ASR, LLM, TTS, and WebRTC handling. [S7]	Compose Azure Speech / avatar with an approved orchestration and model stack.	Ain controls RAG, memory, policy, and evaluation.
Enterprise access	Self-serve and paid routes exist; terms and University approval still required.	Custom voice and avatar are limited-access; Microsoft-managed-customer eligibility is a material dependency. [S5]	Treat access as a discovery spike, not an assumption.
Consent	Custom LiveAvatar requires live or eligible pre-recorded consent depending plan. [S8]	Explicit written permission plus recorded acknowledgement and synthetic disclosure. [S5]	Maintain a provider-independent consent registry.
Control / lock-in	High control in Avatar Only mode but vendor owns visual synthesis.	Potential enterprise alignment but custom access and service constraints apply.	Own source assets, IDs, terms, export / deletion evidence, and fallback.
V1 recommendation	Primary technical spike if approved; use Avatar Only, text fallback, and one consented persona.	Parallel feasibility and institutional-alignment spike.	Do not commit the full product to either until measured.

Important correction to the research note

Azure can support custom video and photo avatars, but real-person use is not an unrestricted single-photo shortcut: consent and limited-access requirements apply. HeyGen LiveAvatar and standard HeyGen video avatars are separate products, and custom LiveAvatar creation is performed through the platform rather than a creation API. [S5][S6][S8]

INTEGRATION CONTRACT

11. API and event surface

Interface	Method / event	Purpose	Key authorization
Session	POST /v1/sessions	Create a scoped conversation session and short-lived media token.	Active user, venture membership, active assigned persona.
Turn	POST /v1/sessions/{id}/turns	Submit text or finalized ASR transcript and start response stream.	Session participant; policy version current.
Stream	WebSocket / WebRTC events	Audio, captions, partial text, interruption, avatar state.	Ephemeral session token; origin and expiry checks.
Evidence	GET /v1/turns/{id}/evidence	Return authorized source and memory references used by the turn.	Same venture / authorized reviewer.
Memory	POST /v1/memory/candidates	Create a candidate durable memory object from a session.	Participant; type-specific approval required.
Memory review	PATCH /v1/memory/{id}	Approve, correct, supersede, pin, or delete eligible memory.	Owner / EIR / operator by object policy.
Escalation	POST /v1/escalations	Create an authorized human handoff packet.	Founder sharing consent plus routing rule.
Persona	POST /v1/personas/{id}/releases	Release a tested persona version.	Named EIR approval and governance gate.
Kill switch	POST /v1/personas/{id}/suspend	Immediately prevent new sessions and terminate active use.	EIR, designated operator, or incident responder.
Audit	event.*	Append immutable security, policy, content, and administrative events.	Service identity; write-only from production paths.

Representative events

```
session.started | turn.transcribed | retrieval.authorized | retrieval.denied | response.validated | response.blocked
| avatar.degraded | memory.proposed | memory.approved | memory.corrected | escalation.created | escalation.resolved |
consent.revoked | persona.suspended | incident.opened
```

Idempotency and correlation

- Every write endpoint accepts an idempotency key; retries must not duplicate memory, escalations, or billing sessions.
- A request ID connects client, gateway, orchestrator, retrieval, model, speech, avatar, policy, and audit events without copying raw content into general telemetry.
- Model and provider callbacks are authenticated, replay-protected, time-bounded, and validated against the expected session and venture scope.

SERVICE QUALITY

12. Nonfunctional requirements and latency budget

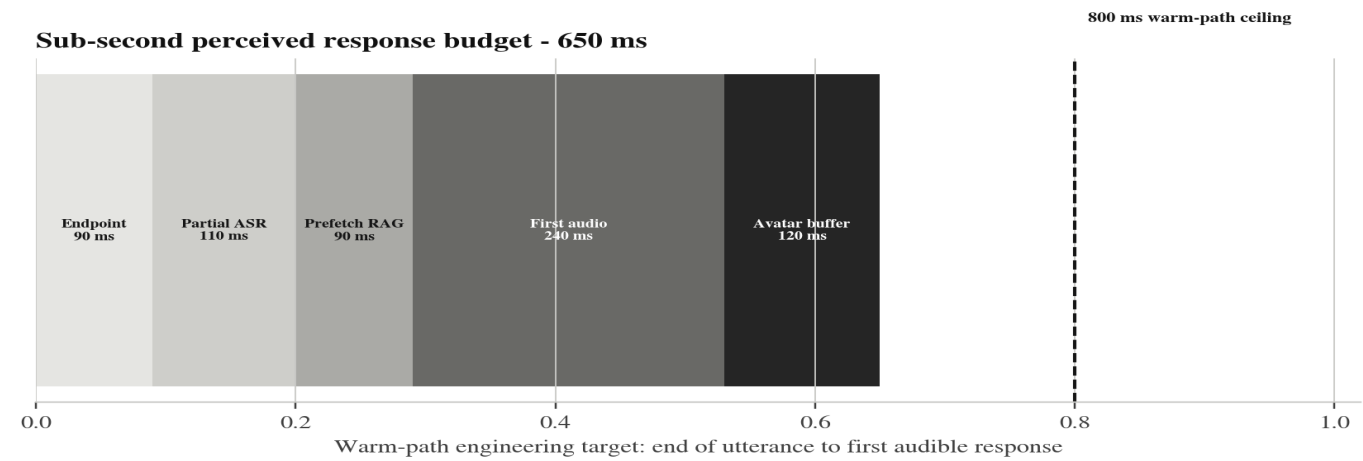
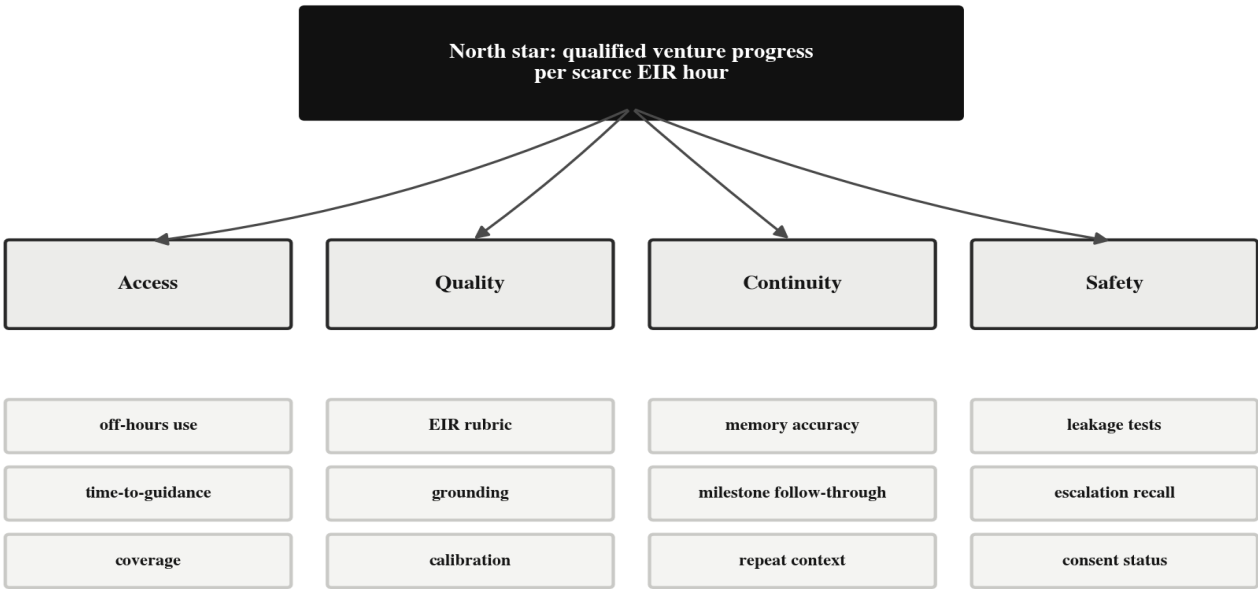


Figure 9. Proposed warm time-to-first-audio budget. Values are engineering targets, not vendor guarantees.

Attribute	Pilot target	Measurement / control
Availability	>= 99.0% during scheduled pilot access windows.	Synthetic sessions; status logging; text fallback excluded only if entire service is unavailable.
First audible response	Warm P50 <= 800 ms; warm P95 <= 1.2 s after end of founder utterance.	Distributed trace across endpointing, partial ASR, retrieval, model / speech, and avatar buffer.
Interruption	Stop generated audio promptly; target <= 200 ms from detected barge-in.	Scripted overlap test and media timestamp analysis.
Grounding	>= 90% of material factual claims source-supported in benchmark.	Claim-to-source validator plus human audit.
Memory accuracy	>= 90% correct recall; 100% retrieved memory traceable to objects.	Seeded multi-session benchmark.
Isolation	0 critical unauthorized disclosures.	Automated authorization suite and prompt-injection red team.
Accessibility	Captions, transcript, keyboard path, text mode, contrast, and screen-reader labels.	Automated checks plus human testing.
Recovery	Text path survives loss of avatar or TTS; no session context corruption.	Fault injection and circuit-breaker tests.
Deletion	Policy-target deletion propagates to primary stores, caches, exports, and eligible vendor assets.	Object inventory plus deletion verification.
Cost visibility	Per-session cost estimated by component and attributable without raw prompt exposure.	Usage meters joined by request / session ID.

OPERATE WHAT CAN BE MEASURED

13. Observability, quality, and evaluation



Leading indicators measure safe use; lagging indicators measure venture progress and EIR leverage.

Figure 10. Product and technical telemetry roll into four balanced dimensions and one human-leverage north star.

Telemetry schema

Signal	Examples	Privacy posture
Technical	Latency by stage, error class, fallback, concurrency, media interruption, provider status.	Metadata only; no raw content required.
Retrieval	Authorized / denied count, sources retrieved, evidence coverage, freshness, conflict count.	Store object identifiers and scope; restrict content snapshots.
Quality	EIR rubric, founder feedback, corrections, hallucination class, escalation appropriateness.	Pseudonymized analytics; raw review samples in restricted workspace.
Safety	Prompt-injection attempt, access denial, policy block, P0 / P1 event, kill-switch exercise.	Security event store with minimum necessary evidence.
Operations	Activation, session minutes, unresolved escalations, persona review backlog, content expiry.	Portfolio aggregates with minimum group thresholds.
Cost	Tokens, ASR / TTS duration, avatar minutes, storage, retries, model route.	Join by session ID; no billing dashboard access to venture content.

Continuous evaluation gates

- Run the gold scenario set on every material prompt, model, retrieval, doctrine, voice, or avatar release.
- Sample low-confidence, low-rated, high-risk, and randomly selected sessions for authorized human review.
- Track score by founder type, venture stage, domain, persona, mode, and language to detect uneven performance.
- Treat a vendor upgrade as a model change requiring regression evidence, even if the API contract is unchanged.
- Separate production monitoring from capstone research analysis and use only approved data for each purpose.

ADVERSARIAL DESIGN

14. Threat model and abuse cases

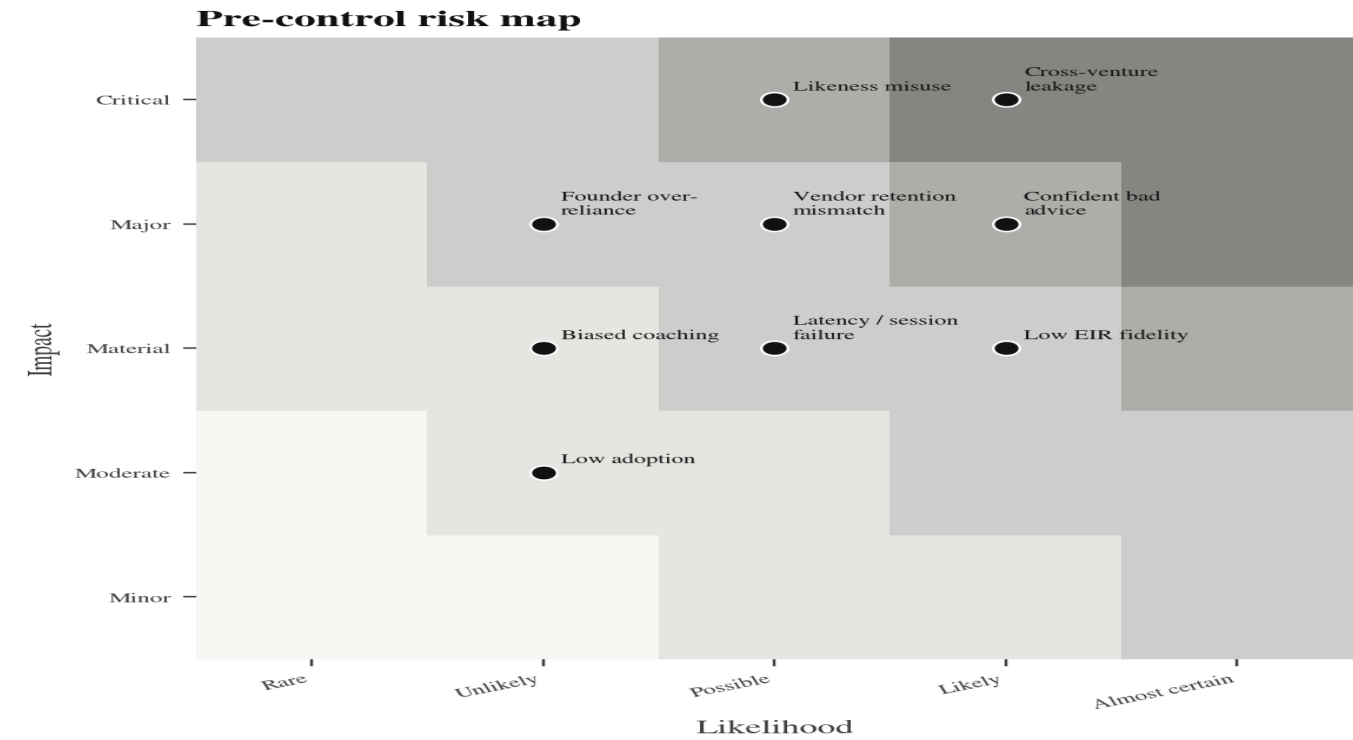


Figure 11. Pre-control risk map guides test depth; it does not replace a formal security assessment.

Threat	Example attack / failure	Required control
Cross-venture exfiltration	Founder asks for another team's customers, deck, lab detail, or conversation.	Server-side scope; row-level security; no global raw vector search; adversarial tests.
Prompt injection	Uploaded document tells the model to ignore policy or reveal system instructions.	Treat content as data; tool allow list; policy outside prompt; output and retrieval validation.
Likeness misuse	Session capture is reused as public endorsement or manipulated clip.	Persistent disclosure, output restrictions, watermark / provenance where feasible, abuse reporting, revocation.
Persona poisoning	Unapproved content shifts doctrine or style.	Signed source inventory, curator approval, checksums, persona versioning, regression tests.
Over-reliance	Founder treats educational guidance as legal, medical, investment, or EIR-approved direction.	Topic classifier, limitations, high-risk escalation, evidence and uncertainty display.
Support-channel leakage	Administrator or vendor support sees raw confidential session content.	Restricted support access, redaction, purpose logging, contractual controls, founder notice where required.
Model memorization / retention	Provider retains prompts or uses data for training contrary to program intent.	Approved terms, minimization, no-training configuration, retention verification, synthetic-data mode.
Denial / cost abuse	Bots or runaway sessions consume avatar and model minutes.	UR authentication, quotas, rate limits, session caps, anomaly detection, circuit breakers.
Stale advice	Resource, competition, regulatory, or funding information changed.	Freshness metadata, source owner, expiry, current web verification, uncertainty and escalation.

REPRODUCIBLE DELIVERY

15. Deployment, environments, and release engineering

Environment	Data	Purpose	Restrictions
Local development	Synthetic users, ventures, sources, voices, and avatars.	Fast development and offline unit tests.	No real EIR biometric assets or confidential venture data.
Integration	Synthetic plus approved public / program test corpus.	Vendor adapters, identity sandbox, end-to-end flows.	No unrestricted production connector access.
Evaluation	Frozen benchmark and expressly authorized pilot data.	Repeatable model, persona, security, and accessibility tests.	Restricted reviewers; immutable release inputs.
Pilot	Approved participant and venture data within the authorized classification.	Real capstone or controlled-pilot use.	Invite-only, monitored, quotas, kill switch, defined support hours.
Production	Program-wide data only after formal approval.	Foundry-scale service.	Production security, operations, contracts, accessibility, incident response, and governance.

Release pipeline

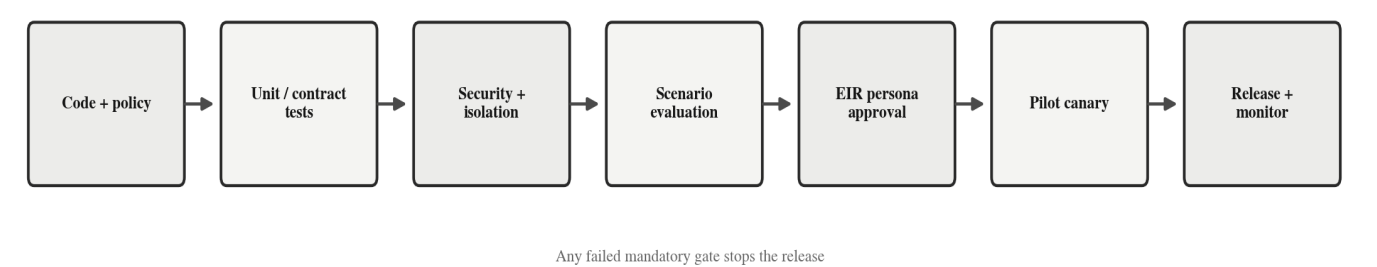


Figure 12. Model, prompt, corpus, policy, persona, speech, and avatar changes travel through the same evidence-bearing release process.

Configuration and secrets

- Keep provider keys, model endpoints, policy versions, and persona asset IDs in managed configuration and secret stores, never source control or browser code.
- Use separate credentials, storage, databases or schemas, callback URLs, and audit sinks by environment.
- Pin model, prompt, corpus, policy, persona, and vendor configuration in each session record to make a response reproducible.
- Automate backups, restore tests, migrations, retention jobs, and access reviews before Foundry-scale operation.

PROOF BEFORE PILOT

16. Verification plan

Test suite	Minimum coverage	Blocking failure
Authorization matrix	All role x venture x persona x data-class combinations, including revoked and expired states.	Any unauthorized read, write, export, stream, or escalation packet.
Cross-venture red team	100+ direct, indirect, prompt-injection, inference, and support-path cases.	Any critical raw or identifying disclosure.
Memory benchmark	Seeded facts, corrections, conflicts, supersession, expiry, and deletion across sessions.	Silent overwrite, untraceable fact, or retrieval after revocation / deletion.
Advisory scenarios	30-50 labeled prompts across stages, business types, and risk classes.	Systematic unsafe direction or median EIR score below gate.
Grounding	Claim-level source coverage, conflicts, stale sources, and insufficient evidence.	Critical unsupported confident claim.
Persona fidelity	Blind EIR scoring of doctrine, questioning, teaching, style, and boundaries.	EIR refuses release or identifies deceptive misrepresentation.
Media / latency	Cold and warm starts, interruptions, poor audio, packet loss, fallback, concurrent sessions.	No usable fallback or session corruption.
Consent lifecycle	Grant, scope change, expiry, revocation, suspension, retirement, deletion evidence.	New or active session after revocation.
Accessibility	Keyboard, captions, transcript, text-only, focus, contrast, zoom, screen reader.	Core founder job unavailable without audio / video.
Incident drill	Leakage allegation, compromised account, harmful response, vendor outage, likeness abuse.	Cannot contain, preserve evidence, notify owner, or suspend affected paths.

Required evidence bundle

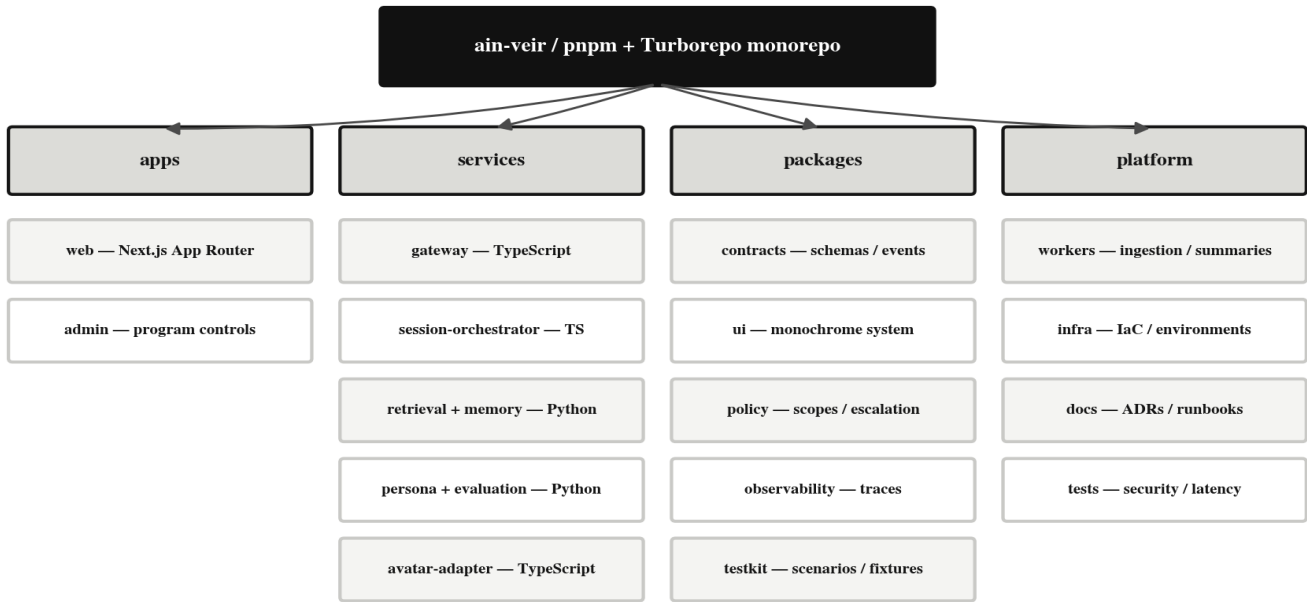
- Machine-readable test results, benchmark version, code commit, configuration manifest, and evaluator identities / roles.
- Representative failed cases and known limitations, not only aggregate success rates.
- Architecture and data-flow review, vendor settings, consent status, access matrix, and retention map.
- A signed or otherwise attributable release decision by the technical owner, program owner, and relevant EIR approver.

BUILD PROFILE

17. Language standard and repository blueprint

Codebase decision

Use TypeScript as the product language for the web shell, BFF, real-time session gateway, orchestration, provider adapters, and shared contracts. Use Python where its data and AI ecosystem is strongest: retrieval, ingestion, evaluation, memory consolidation, and experiments. Introduce Rust only after profiling identifies a media or parsing hot spot that Node and native libraries cannot meet.



One deployable product first; split services only where scaling, isolation, or language ownership requires it.

Figure 13. Concrete monorepo boundary. Directory names are implementation guidance, not placeholders.

Layer	Primary language / framework	Reason and constraint
Product UI	TypeScript; Next.js App Router; React; Tailwind or CSS variables; Storybook	One typed route and component model for founder, EIR, program, and admin experiences. [S19]
Realtime control	TypeScript; Node.js; LiveKit SDK; WebSocket / WebRTC events	Low-overhead event streaming and contract reuse with the client; regional stateless workers. [S21]
AI / data services	Python 3.12; FastAPI; Pydantic; SQLAlchemy; evaluation tooling	Mature retrieval, document, model, and analytics ecosystem; keep APIs schema-first.
Authoritative data	PostgreSQL; pgvector; Postgres FTS; row-level security	One transactional authority for identity mappings, assignments, memory, evidence, versions, and hybrid retrieval. [S22]
Optional optimization	Rust library or sidecar only after measured need	Suitable for codec, parsing, or CPU-bound hot spots; prohibited as speculative V1 complexity.

HOT PATH VERSUS DURABLE WORK

18. Services, layers, and orchestration

Runtime topology: interactive path separated from durable work

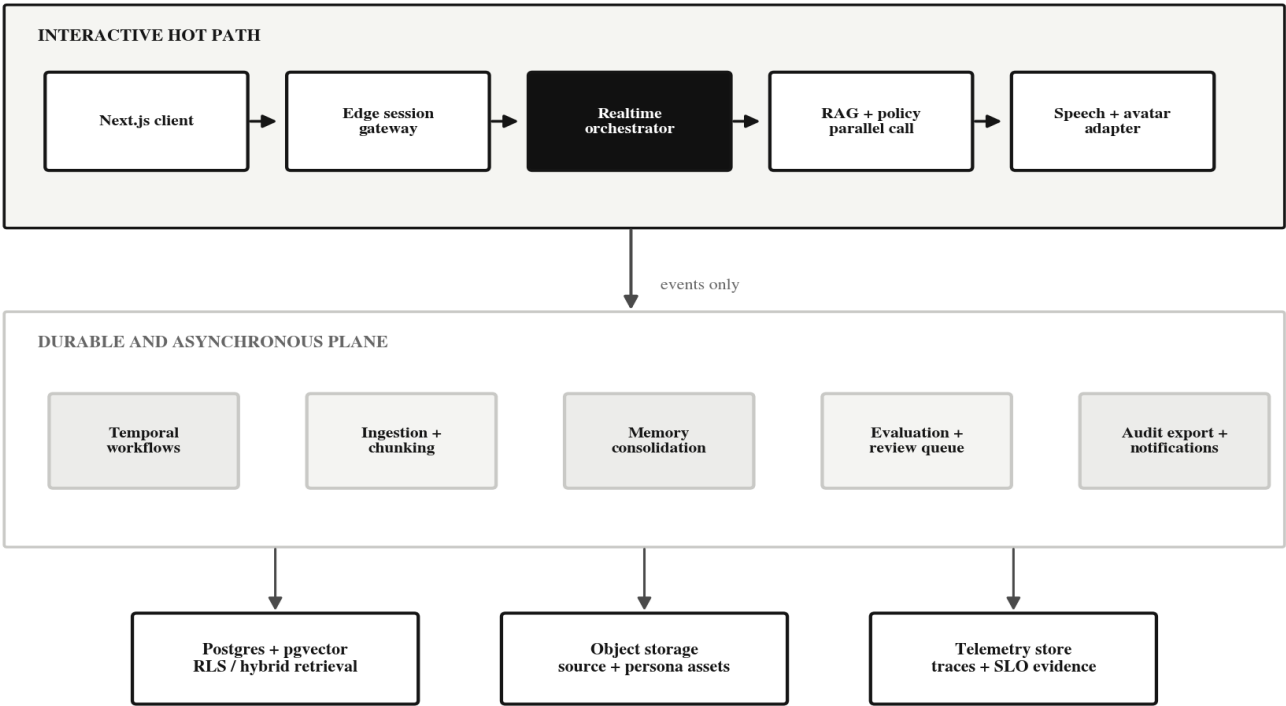


Figure 14. Realtime interaction remains a narrow event path; Temporal coordinates durable work without adding turn latency. [S23]

Runtime unit	State and scaling	Owens / explicitly does not own
web	Edge-cached shell; server-rendered route frames; client WebRTC state	Owens presentation and local media controls; no secrets, policy authority, or raw cross-venture search.
gateway	Stateless regional instances; scale on sessions and requests	Owens auth context, BFF aggregation, quotas, room tokens; does not generate advice.
session-orchestrator	Warm regional workers; ephemeral turn state mirrored to session store	Owens endpointing, interruption, stream sequencing, mode and tool plan; does not persist canonical memory directly.
retrieval	Stateless Python workers; scale on concurrent queries	Owens scoped hybrid search, reranking, evidence packs; cannot bypass RLS or assignment policy.
memory	Transactional service over Postgres	Owens typed proposals, approval, correction, supersession, expiry, deletion; no silent fact creation.
persona	Versioned registry and release API	Owens doctrine, style, consent references, test status, activation; avatar vendor is not authoritative.
avatar-adapter	Stateless TypeScript; one circuit breaker per provider / region	Owens normalized create / stream / interrupt / close contract; never receives full venture corpus.
workflow workers	Temporal activities; independently scalable	Own ingestion, reviews, exports, recertification, deletion and notifications; excluded from live speech critical path.

SUB-SECOND PERCEIVED RESPONSE

19. Streaming latency and global edge

Global real-time media with regional warm workers

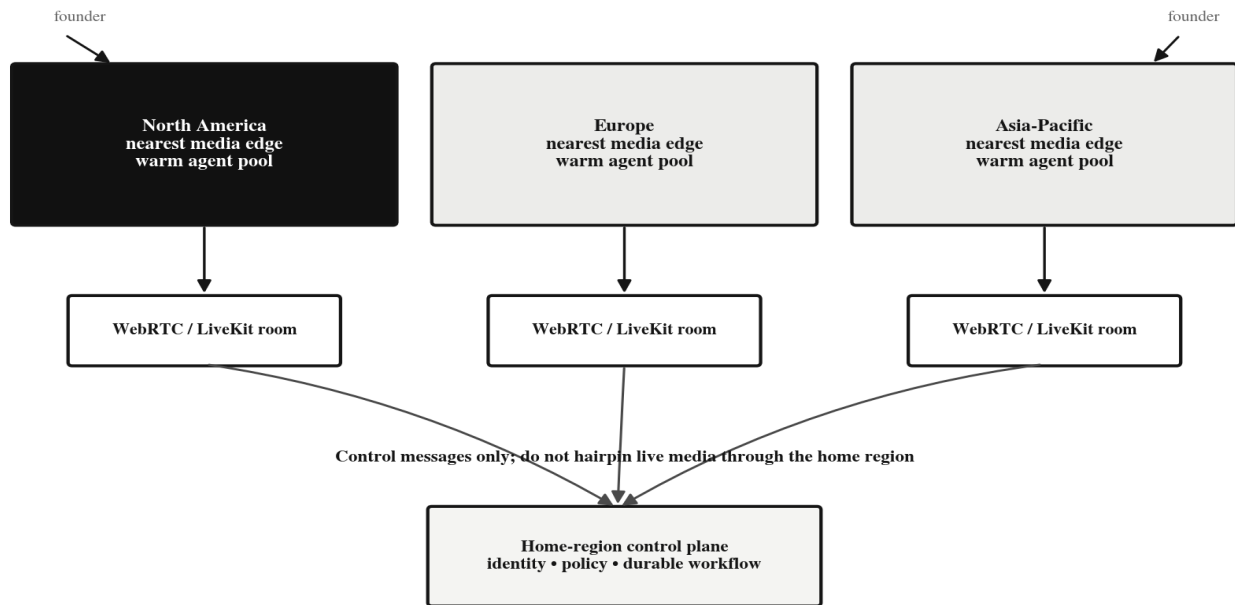


Figure 15. Founders join the nearest media edge and a warm regional agent; control-plane messages can return home without hairpinning live media. [S21]

Literal zero latency is physically impossible. The product target is a response that **feels immediate**: direct WebRTC media, partial speech recognition, early endpointing, venture-memory prefetch, incremental model / speech output, and a small avatar jitter buffer. OpenAI Realtime supports WebRTC and WebSocket transports; LiveKit provides WebRTC rooms and global-edge patterns. [S20][S21]

CRITICAL-PATH ENGINEERING

19A. Latency budget and tuning rules

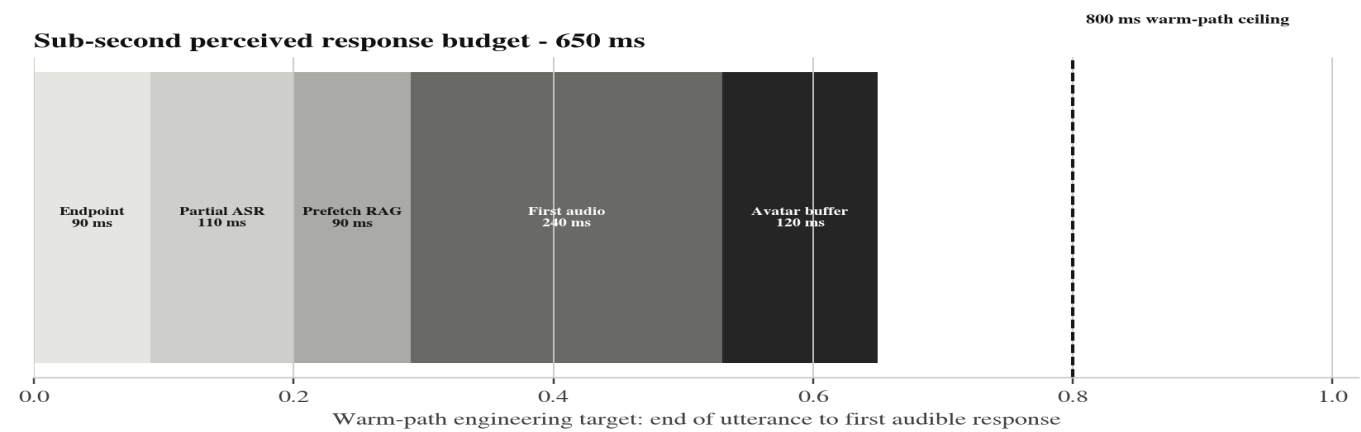


Figure 16. Warm engineering budget totals 650 ms, leaving 150 ms of the 800 ms ceiling for variance.

Signal	Target	Engineering action
First audible response	Warm P50 <= 800 ms; P95 <= 1.2 s	Regional warm pools; partial ASR; speculative scope-safe prefetch; incremental audio.
Barge-in	Generated audio stops <= 200 ms after detection	Client voice activity event; cancel turn; flush speech / avatar buffers; idempotent resume.
Media path	Typical glass-to-glass contribution < 150 ms where network permits	Nearest LiveKit edge; ICE / TURN fallback; no home-region hairpin.
Cold / degraded	Reported separately; never hidden in warm averages	Pre-warm based on schedule; text-first fallback; expose provider and region in traces.

Critical-path implementation rules

- Open the media room and regional agent before the founder begins a turn; avoid per-turn container and provider startup.
- Run partial ASR, intent / risk classification, and venture-memory prefetch concurrently, then reconcile when endpointing finalizes.
- Stream only policy-approved response units; cancellation propagates to model, speech, avatar, captions, and billing state.
- Bound context, tool count, first-sentence length, and avatar buffer; defer recap, durable memory, analytics, and notifications.
- Publish P50, P95, cold-start, region, fallback, packet-loss, and provider results separately so averages cannot hide degraded paths.

SCOPED INTELLIGENCE

20. RAG, venture memory, and evidence

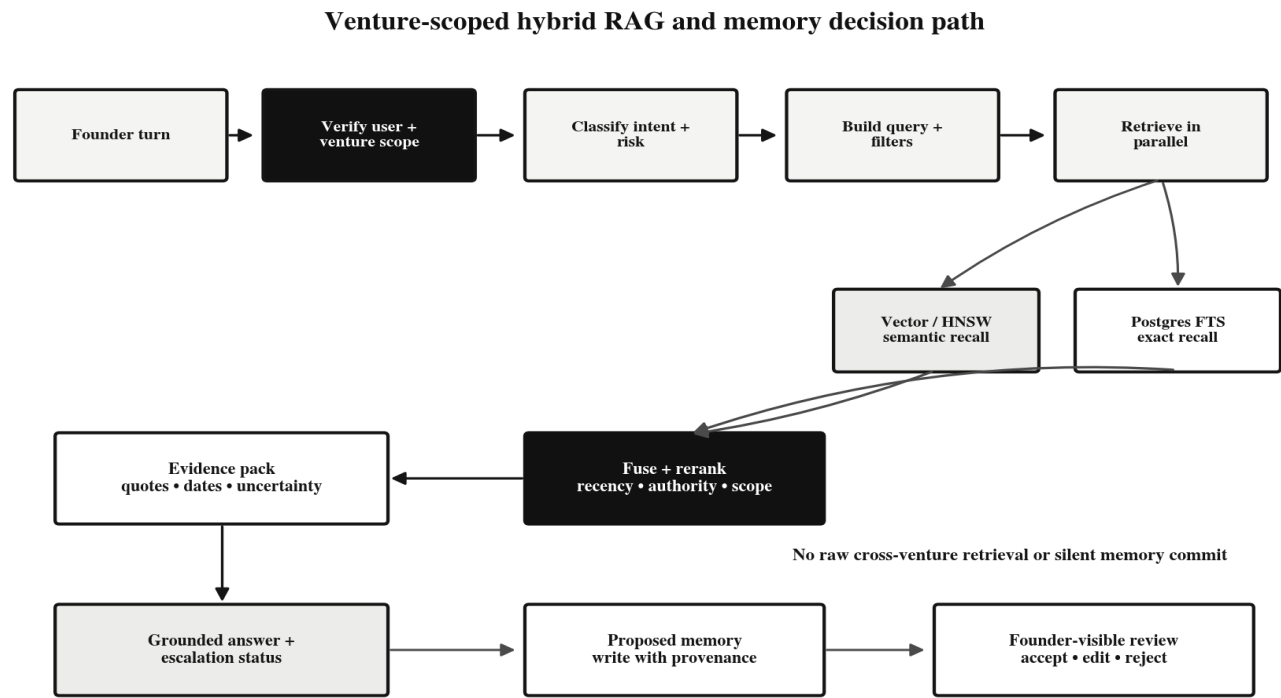


Figure 17. Authorization precedes retrieval; lexical and semantic recall are fused, reranked, and converted into a visible evidence pack before generation. [S22]

Memory tier	Typical objects	Runtime priority and protection
Current session	Turn transcript, active goal, interruption, selected artifact	Highest immediacy; short retention; participant scope.
Venture private	Validated facts, assumptions, decisions, experiments, milestones, files	Exact venture only; RLS; versioned corrections; material writes reviewed.
EIR doctrine	Frameworks, questions, red lines, cases, referrals, teaching patterns	Only active approved persona release; never grants venture access.
Ain shared	Foundry steps, resources, competitions, mentors, routing and calendars	Program policy and freshness filters; named source owner. [S16][S17][S18]
Reviewed patterns	De-identified lessons explicitly approved for reuse	Lowest priority; minimum-group and de-identification gates; no raw transcript pooling.

Ranking and answer rule

Use a weighted fusion of semantic relevance, lexical match, source authority, freshness, venture-stage fit, and contradiction penalty. A high similarity score never overrides access scope. Each material claim must resolve to an evidence object or be labeled as inference, hypothesis, or recommendation; insufficient coverage narrows the answer or triggers escalation.

BOUNDED PROFESSIONAL ALIGNMENT

21. Persona, emotion, style, and interaction fidelity

Persona fidelity is a governed behavior stack—not emotion mind-reading

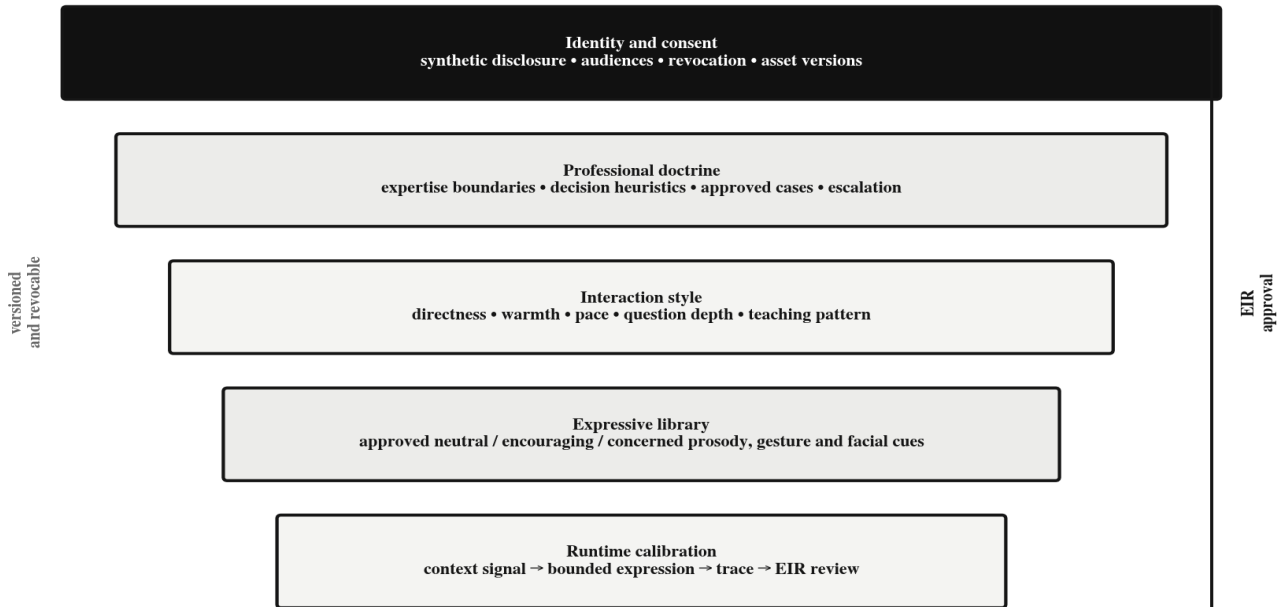


Figure 18. The system captures consented professional behavior and approved expressive cues; it does not claim to infer or reproduce an EIR's private emotions.

Dimension	Captured from	Runtime representation	Evaluation
Expertise	Public profile plus EIR-approved interviews and materials	Topics, boundaries, source preferences, escalation categories	Scenario correctness and boundary adherence
Reasoning style	Decision cases and think-aloud interviews	Question order, evidence threshold, trade-off and counterexample patterns	Blind EIR rubric against gold cases
Interaction style	Consented session exemplars and explicit style card	Directness, warmth, pace, vocabulary, question depth, teaching mode	Pairwise fidelity review; founder clarity / trust
Expressive affect	Provider-compliant video / voice capture plus approved cue library	Bounded neutral, encouraging, concerned, reflective, and celebratory delivery	EIR approval; detect overacting, mismatch, or manipulation
Identity	Written and provider-recorded consent where required	Persistent synthetic label, persona release ID, revocation state	No active session after expiry or revocation

Product wording
Promise 'professionally aligned interaction'—not emotional cloning. The vEIR may select an approved expression from observable session context, but it must never state that the human EIR personally feels, believes, or endorses a generated response.

MINIMAL, ROBUST, EVIDENCE-FIRST

22. Frontend shell and visual system

Founder session shell — muted black, gray, and evidence-first

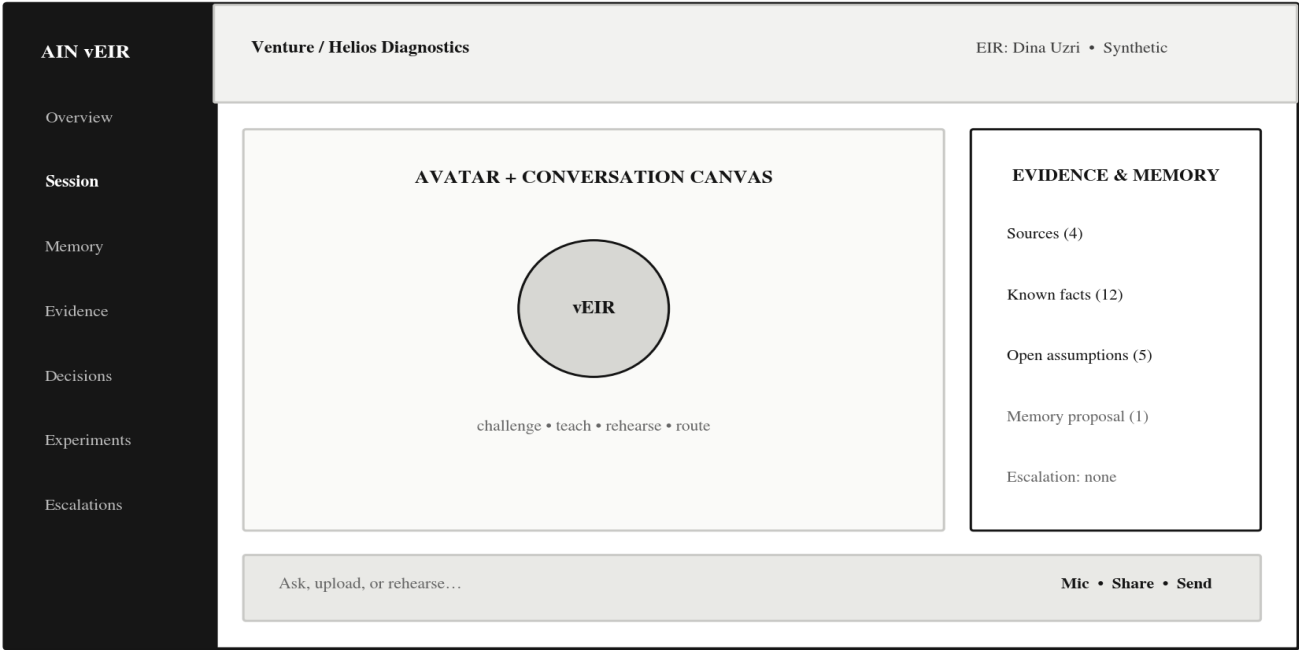


Figure 19. Session shell keeps venture context, synthetic identity, evidence, memory, controls, and escalation visible without turning the avatar into the entire product.

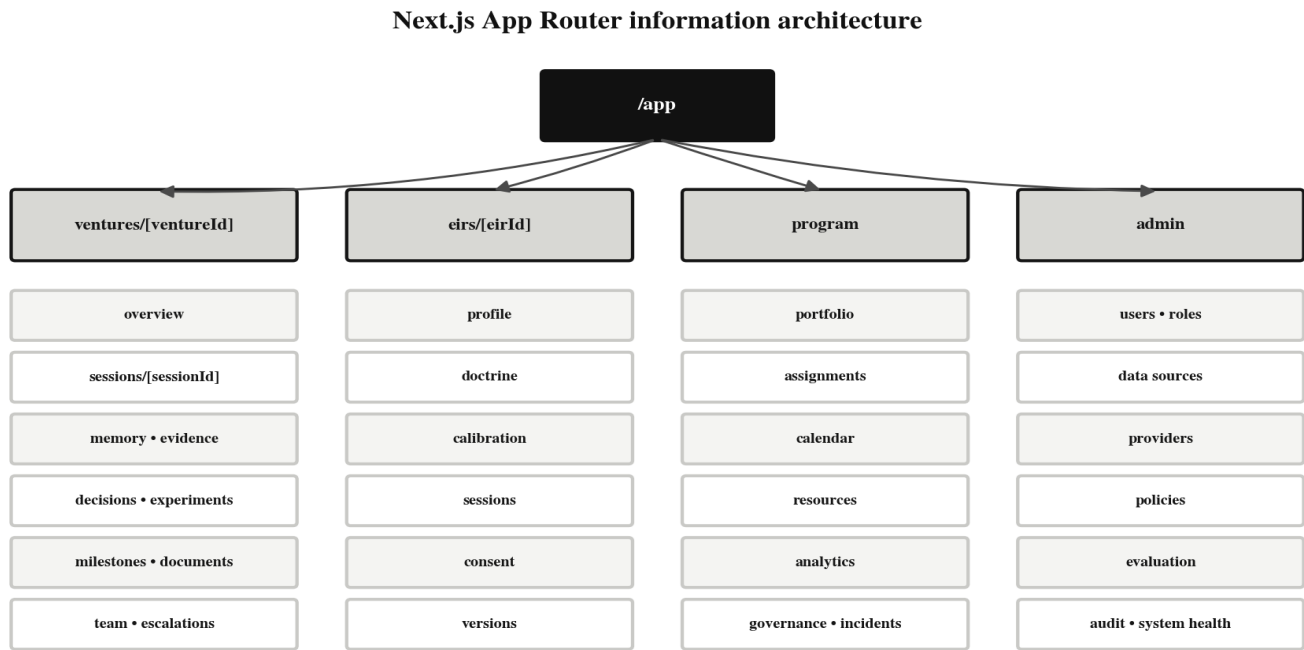
Region	Persistent responsibility	Responsive behavior
Navigation rail	Venture workspace and major routes; active route; role-safe switcher	Collapses to icon rail / drawer; never loses venture identity.
Context bar	Venture, stage, assigned EIR, persona release, synthetic label, session health	Wraps to two rows; critical disclosure remains visible.
Conversation canvas	Avatar or text mode, transcript, coaching mode, live captions, interruption state	Video yields to transcript and controls on smaller screens.
Evidence / memory inspector	Sources, known facts, assumptions, memory candidates, contradictions, escalation	Becomes tabbed sheet; keyboard reachable; badges are not color-only.
Control dock	Mic, camera, upload, rehearsal target, send, stop, share / escalate	Sticky; large targets; explicit destructive and sharing confirmations.

Standardized visual tokens

UI canvas #0B0B0B; raised surface #151515; border #303030; primary text #F4F4F2; secondary text #A6A6A3; focus ring #FFFFFF. Use Times-compatible editorial typography only in documents; use a legible system sans-serif in the app. Status must combine label, icon, and shape so safety meaning never depends on color. Respect reduced motion and offer text-only mode.

PAGES WITHIN PAGES

23. Routes, layouts, and component ownership



Nested layouts preserve venture context, role controls, active EIR, session state, and evidence panels.

Figure 20. Nested App Router layouts preserve venture and role context while child pages stream independently. [S19]

Layout / component	Owns	Must never own
RootLayout / AppShell	Global theme, authenticated shell, error / loading frames, command palette	Venture authorization or secrets
VentureLayout	Server-resolved venture membership, venture header, left nav, evidence / memory panel state	Client-supplied scope as authority
SessionPage	ConversationCanvas, AvatarStage, transcript, ControlDock, live status	Canonical memory writes or unvalidated provider callbacks
EvidenceDrawer	Claim-to-source links, dates, authority, conflicts, uncertainty	Hidden global search or cross-venture snippets
MemoryInspector	Typed objects, provenance, candidate diff, accept / edit / reject	Silent auto-approval of material memory
EIRCalibrationStudio	Doctrine diff, scenario runner, sample review, release / suspend controls	Venture access absent assignment and purpose
ProgramPortfolio	Minimum-safe aggregates, queues, utilization, incidents	Raw founder transcripts by default

PORTABLE PRODUCTION TOPOLOGY

24. Deployment, provider adapters, and resilience

Concern	Selected pattern	Failure boundary
Global entry	DNS / edge routing to the nearest approved region; signed short-lived session token	Region unavailable → next allowed region or text-only home path
Media	LiveKit Cloud global mesh for fastest pilot path, or approved multi-region self-hosted LiveKit	ICE / TURN and reconnect; avatar loss never removes text or evidence. [S21]
Avatar	Adapter contract: createSession, pushAudio / text, interrupt, health, close	HeyGen LiveAvatar Lite / Avatar Only, Azure avatar, or Tavus evaluated behind the same contract. [S5][S6][S7][S8][S25][S26]
Reasoning / speech	Provider-neutral streaming contract; native speech-to-speech when approved, otherwise partial ASR → model → TTS	Timeout and circuit breaker per model / region; no unapproved voice substitution. [S20]
State	Postgres primary authority; Redis only for ephemeral room / turn coordination; object storage for source assets	Cache loss rehydrates; database unavailability blocks writes and degrades safely
Durable jobs	Temporal namespace per environment; retry policies, idempotent activities, human approval signals	Workflow outage delays background work but does not enter live media path. [S23]
Edge coordination	Optional Durable Object or equivalent for lightweight session-presence coordination	Never store canonical venture memory or route raw media through the coordinator. [S24]

Provider adapter contract

```
AvatarProvider = { capabilities(): ProviderCapabilities; createSession(scope, personaRelease): SessionHandle;
streamInput(sessionId, approvedChunk): AsyncEventStream; interrupt(sessionId): Ack; close(sessionId): UsageReceipt;
deletePersonaAsset(assetId): DeletionEvidence }
```

Resilience rule

Each external call has an explicit timeout, circuit breaker, request / session ID, data-minimization map, retention expectation, and user-visible fallback. Retries are bounded and idempotent; a retry cannot duplicate spoken output, media billing sessions, memory writes, or escalations.

CODE TO CONTROLLED PILOT

25. Ninety-day implementation sequence

Sprint	Build outcome	Exit evidence
1 — Foundations	Monorepo, CI, typed contracts, synthetic fixtures, Next.js shell, Postgres schema, identity stub	Local one-command setup; schema / contract tests; no secrets in client or repository
2 — Scoped knowledge	Ingestion worker, chunk provenance, pgvector + FTS, RLS, EvidenceDrawer	Positive / negative authorization matrix; source-linked answer benchmark
3 — Venture memory	Typed memory API, proposals, corrections, supersession, MemoryInspector	Multi-session recall and deletion tests; no silent material commits
4 — Realtime core	LiveKit room, endpointing, partial ASR, stream orchestrator, barge-in, text fallback	Warm P50 / P95 trace; <= 200 ms interruption target; degraded-mode drill
5 — Persona + avatar	Consent registry, persona releases, style / doctrine cards, provider adapter, disclosure	EIR-approved representative sessions; suspension and revocation test
6 — Program operations	Assignments, escalation queue, EIR calibration, admin policy, audit, cost traces	End-to-end founder → vEIR → human handoff with minimum-necessary packet
7 — Pilot hardening	Threat tests, accessibility, failure injection, privacy review, operational runbooks	All blocking gates pass on frozen release manifest
8 — Controlled cohort	8–12 ventures, repeated sessions, founder and EIR evaluation	Proceed / revise / stop recommendation with evidence and known limitations

Definition of a pilot-candidate release

- One consenting EIR and each participating venture have active, time-bounded assignment and data-scope records.
- The full text path works without avatar video; every material claim exposes evidence or an uncertainty label.
- Warm latency, barge-in, cross-venture isolation, memory accuracy, escalation recall, accessibility, and consent lifecycle meet agreed gates.
- The release manifest pins code commit, schema, model, prompt, corpus, policy, persona, voice, avatar provider, region, and evaluation versions.
- Named technical, program, privacy / security, and EIR owners sign the pilot decision; a kill switch and incident runbook are tested.

DECISIONS MADE EXPLICIT

26. Architecture decision records and handoff

ADR	Decision	Status / revisit trigger
ADR-001	Own vEIR orchestration, memory, evidence, and policy; abstract the avatar body.	Proposed - revisit after two vendor spikes.
ADR-002	Use one testable orchestrator with deterministic validators in V1.	Proposed - revisit only if a bounded multi-agent task materially outperforms.
ADR-003	Use explicit typed memory with approval, provenance, correction, and supersession.	Proposed - foundational; do not replace with opaque chat history.
ADR-004	Enforce venture isolation in service and database controls, not prompts.	Required - deployment blocker.
ADR-005	Keep persona expertise, judgment, style, voice, and visual likeness separable.	Required - consent, portability, and safety.
ADR-006	Begin with public, synthetic, or expressly authorized low-risk data.	Required until the applicable University approval expands scope.
ADR-007	Text / caption mode is a complete fallback, not a secondary afterthought.	Required - accessibility and resilience.
ADR-008	High-risk topics produce educational framing and structured human escalation.	Required - product boundary.
ADR-009	TypeScript is the primary product / realtime language; Python owns AI and data workloads.	Proposed - revisit only with measured team or runtime evidence.
ADR-010	Use WebRTC / LiveKit regional rooms and keep durable workflows outside the audio hot path.	Proposed - validate provider, region, and latency measurements.
ADR-011	Use Postgres hybrid retrieval with RLS before ranking and explicit memory approval.	Required - isolation and audit foundation.
ADR-012	Use nested Next.js layouts with a persistent evidence-and-memory session shell.	Proposed - validate accessibility and founder usability.

Implementation handoff checklist

- Repository, local setup, architecture diagrams, environment inventory, dependency and license list.
- API specification, event catalog, data dictionary, migrations, seed data, and synthetic-data generator.
- Persona card schema, consent schema, benchmark scenarios, evaluation rubrics, and release gates.
- Threat model, authorization matrix, red-team suite, accessibility results, incident and rollback runbooks.
- Vendor due-diligence record, terms / configuration mapping, deletion path, cost model, and exit plan.

TECHNICAL EVIDENCE BASIS

27. Source register

Primary sources are favored for program facts, product documentation, University policy, and technical implementation patterns. All provider behavior, regional availability, consent workflow, pricing, retention, and enterprise eligibility must be revalidated during procurement and before each release.

S1 Ain Center for Entrepreneurship and Innovation. Analytics Projects course sponsor briefing with Professor Roberto Colangelo. Video recorded August 31, 2026. User-provided source.

S2 University of Rochester News Center. Student entrepreneurs learn business by starting one. July 29, 2026.
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S3 Ain Center for Entrepreneurship and Innovation. Entrepreneurs in Residence. Accessed September 2026. <https://www.rochester.edu/aincenter/entrepreneurs-in-residence/>

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<https://www.rochester.edu/university-research/about/committees/ai-research-committee/responsible-use-gen-ai/>

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S16 Ain Center for Entrepreneurship and Innovation. Ain Foundry Program. Accessed September 2026. <https://www.rochester.edu/aincenter/ain-foundry-program/>

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S22 pgvector. Open-source vector similarity search for Postgres, including exact, HNSW, IVFFlat, and hybrid-search patterns. Accessed September 2026.
<https://github.com/pgvector/pgvector>

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